

Analysis of Database Programming Technology in Computer Software Engineering

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Abstract: In the process of rapid development of information technology, the scale of the information industry is continuously expanding, which creates very favorable conditions for the construction and development of computer software engineering. In computer software engineering, programmers need to enhance the application value of programming technology, so as to ensure that they can meet the various needs of users. This requires programmers to study and analyze database programming technology based on computer software engineering. At the same time, the research is carried out from the computer software engineering technology form, analyzes various problems existing in the application process of database programming technology, and puts forward the key points of database programming technology application. Only in this way can we ensure that the functional characteristics of the computer database can be fully exerted, and promote the further development of my country's information industry.

Keywords: Computer Software; Database Programming Technology; Application Analysis

I. Introduction

In the process of computer software engineering construction, database programming technology is an important part of it. Besides, It will affect the computing power of the computer to a certain extent. Therefore, we need to pay attention to the research and analysis of database programming technology. In the current rapid development of the social economy, the demand for the application of computer software engineering is increasing. In order to be able to give full play to the design value of computer software engineering, we not only need to attach importance to the construction of software engineering, but also to effectively integrate with the problems of traditional software computers. Only in this way can the computer software engineering projects be fully integrated to improve the innovation and application value of the computer system. In the development process of computer software engineering, we should improve the value of software use according to engineering construction methods. At the same time, we must integrate database programming design technology and software development engineering and other technologies, so as to provide effective support for the innovative development of modern software systems and database programming technology.

II. Computer Software Technology Overview

In software engineering, staff not only need to strengthen the development of software resources, but also make gradual changes to the logic content of high-level processing, and improve and optimize the goals in the implementation process

of low-level processing. At the same time, we must promote the optimization and development of computer software technology. In the process of large-scale software development, the form of computer engineering technology contains many factors. Staff, technology, and cost are all components of the form of technology. For some large software projects, the complexity of the conceptual form is relatively high. However, in the context of software engineering, the use of software products and software engineering for resource optimization can fully consider the user's use needs and complete the software system design work. In addition, the theoretical level of the software computer is improved to ensure the effectiveness of engineering design. In the application of computer software technology, the engineering technology form of software design is mainly manifested in the following aspects: first, the life cycle. That is to say, in the process of software project design, we need to proceed from the aspects of software design, specification maintenance, and coding to ensure that the project design has a strong periodicity. At the same time to meet the basic needs of customer design. In addition, the purpose of software design should be clarified during the analysis of the role of customer specifications, and the engineering procedures of the computer system should be further optimized. Second, the specifications. In the process of software engineering development technology analysis, we must pay attention to software development. Besides, we need to strengthen the input and output required by users based on the design of the computer system. In this way can the use value of computer data be improved, and the application level of computer software program engineering technology can be improved at the same time. Third, strengthen the analysis of software technology. In this process, programmers need to focus on the effective use of computer technology and the management of hardware devices, so that they can conduct research and analysis of device specifications and device content and complete software system development work. Simply in this way can we optimize and improve the core goals of the system design, ensure that the system design goals can be achieved, and ensure the design effect [1].

III. Problems in Database Programming Technology in Computer Software Engineering

There are still some problems in the application of database programming technology in computer software engineering. These problems are mainly manifested in the following aspects: first, the cost is relatively high. In the process of using database programming technology based on computer software engineering, we must pay attention to computer software engineering design work. However, the traditional computer software technology is relatively

backward, and the loss of manpower, material resources and financial resources is also relatively serious. If these problems cannot be solved in time, the cost of software systems will increase. Moreover, in the process of assembling computer software, we may also effectively combine the use of foreign computers to complete the product purchase process. In this way, we can not only further ensure the application value of software product design, but also optimize the value of capital circulation, thereby providing an accurate reference for the cost of modern engineering. In the application of computer software equipment, the price of software products is relatively high, in this case, it will affect the company's capital flow to a certain extent. In addition, this will lead to a reduction in the operating effect of the computer equipment, which will have a certain impact on the safety of the computer and the safety of the computer. Second, the load loss is relatively large. For the analysis of the use of computers, we will find that in the development of new technologies, we can use the analysis of user system operating conditions, and at the same time improve the design level of computer software engineering in conjunction with the software design process. Otherwise, this will also bring challenges to software engineering, and in the development of the computer industry, we need to comprehensively analyze the specific conditions of social operation and development. In this way can we effectively develop traditional computer resources, thereby enhancing the use value of computers. In these processes, the load loss of computer software engineering will be increased to some extent. Third, the system work efficiency is relatively low. In the traditional computer software engineering design process, the engineering structure and engineering projects are not complete enough, resulting in a relatively low level of computer software quality operation, which will affect the operating efficiency of the software. For example, in the process of management and integration of database resources, computer software cannot meet the requirements for safe operation, so that there are problems with the storage method and processing effect, which has a certain impact on the stability of the office system, resulting in a reduction in user experience [2].

IV. Establish Computer Software Engineering Database

In the context of computer software engineering, when making full use of database programming technology, we must build a more powerful database. Therefore, we need to proceed from the following aspects: First, we must strengthen the research and analysis of the functional characteristics of the database system. When building computer software engineering, we must build a basic framework, and we must fully consider the operating conditions in the network environment to ensure that the database construction work based on computer software engineering is scientific and reasonable. At the same time, in the process of building the database, we have to evaluate the application effects of different components, and we must study and analyze the factors that affect the application effect. Because in this way

can the database application based on computer software engineering achieve the expected design goals. Second, when establishing the database, we must clarify the application direction of the database before we can implement effective research work. When designing the basic framework, we must improve the function according to the specific situation and ensure the effectiveness of the basic framework design. This can make the database have good functional characteristics, improve the information security of computer software engineering during operation, and make the database effectively respond to network viruses. At the same time, we must speed up the update rate of the database in the practical application process. Third, when building the database, we not only need to conduct a more in-depth discussion and analysis of the specific situation of computer software engineering, but also need to focus on the technical difficulties in the process of building and strengthen the coordination of different software functions. This can effectively reduce the possibility of technical problems in the database creation process. In addition, when we practice computer software engineering, we must also pay attention to the efficiency of the use of database resources, we must choose the appropriate program assembly language to provide more support for the establishment of the database. In the long-term practice of the database, programmers must actively explore ways to establish a database based on computer software engineering. This can further improve the rationality and scientificity of database establishment, and lay a good foundation for the further development of database programming technology [3].

V. Application of Database Programming Technology in Computer Software Engineering

A. Choose Object-oriented Database Storage Mode

In the process of selecting the database mode, we need to complete the storage function of selecting the database after partitioning. The database storage mode mainly includes three modes: internal mode, external mode and conceptual mode, as shown in Figure 1. Selecting the database storage mode is also the process of defining user access rights. When applying software, in order to ensure the security of internal information, we need to define the user's access rights. This can ensure that different levels of user login interface are different, and can ensure the security of database information. The rationality of file access based on file type selection is relatively high. In order to achieve this function, programmers need to complete the database selection according to the type of information stored in the process of writing the program to prevent deeper problems from affecting the rationality of database storage mode selection. Using scientific and reasonable database storage mode, database resources can be optimized reasonably, and the application effect of database storage can be guaranteed. In the process of using different resources, the program mobilizes the internal information of the database according to the instructions received, which can greatly improve the utilization efficiency of data resources.

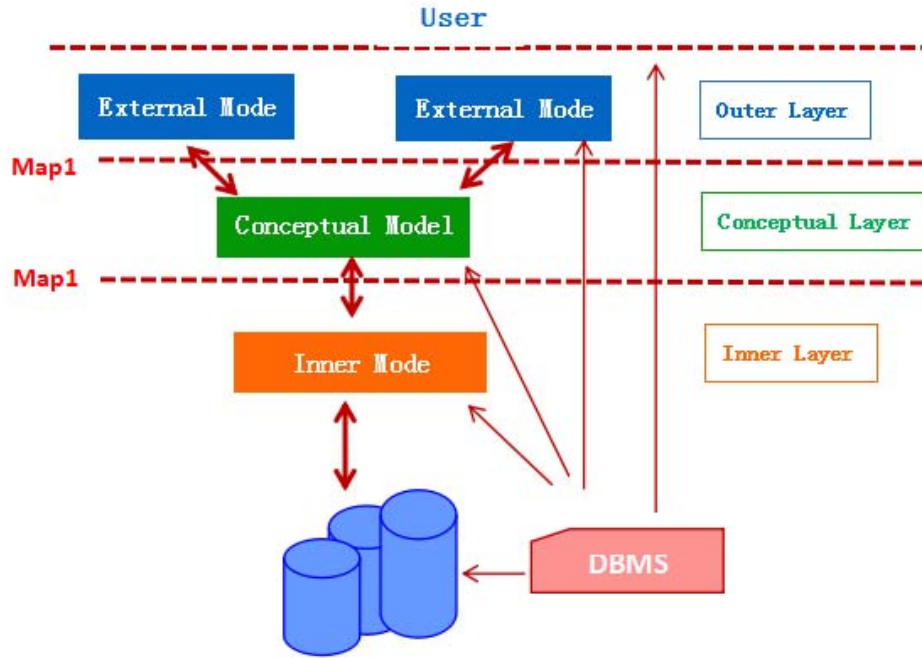


Figure 1 Composition of Database Storage Mode

B. Optimize Database Performance

The application of database programming technology based on computer software engineering needs to optimize database performance. The optimization and application of database performance is shown in the following aspects: First, we can use the performance optimization function of database programming technology to select the appropriate program development language to complete the logic editing according to our actual needs. In this way, the functional architecture of the database can be effectively set, and the functional partition can be completed according to the utilization rate of each module, which can effectively improve the utilization efficiency of the database resources and ensure the optimization effect of the database performance. Second, by using database programming technology in the process of applying it rationally, we can make full use of the advantages of database programming technology and partition construction while determining the database file type. This can ensure that the database application functions in different regions can be fully exerted, and reduce the data conflicts in the database application process. Third, in the process of using the programming technology, programmers can provide database applications with more diversified services based on the user's independent selection needs, and further improve the efficiency of database resource utilization [4].

C. Encrypt and Protect Database Files

The encryption and protection of database files during the application of database programming technology based on computer software engineering are mainly for basic information. In this part of the information encryption

protection process, it is mainly to protect the user's personal privacy. Because, if personal privacy is leaked, it will cause serious adverse consequences. Therefore, in the development of database file encryption protection, we must set the encryption protection level according to the importance of different information. We can use the login password and password encryption to complete the file encryption protection. When logging into the database file, we need to enter the corresponding encryption key. Now the only way can the staff ensure that the encryption measures are researched and analyzed according to common problems, and the files are prevented from being paralyzed during the use of the database by the application of network viruses during the application process. When selecting the data information during the period, you need to use file encryption between different systems to select. If there is a conflict in function, we can optimize through the system frame structure to improve the optimization effect and ensure that the function optimization between various systems can achieve the expected goal.

D. Development Programming Technology

Based on the selection status of database programming technology, when analyzing the stability of computer software engineering, programmers need to strengthen the optimization of the system operating status and resource optimization issues. Moreover, they must keep abreast of the restrictive problems that exist during the operation of the system, so that the value of the computer system's operation can be demonstrated. When researching and analyzing system function problems, we need to observe the stability of the system operation based on the restrictive problems in the programming work. At the same time, we must make full use of the data transmission

information based on the data transmission problem, so that the data control process can be optimized, thereby increasing the use value of the software function and ensuring the effectiveness of the data processing results. When testing the program, there may be a problem with restrictive technology choices simultaneously. In this case, programmers need to complete software stability performance testing according to the specific conditions of software development, and optimize the use value of programming technology. Meanwhile, it provides effective guarantee for the stability of system operation. Besides, in the analysis of software engineering stability, software stability and comprehensive programming techniques can be studied and analyzed to optimize programming methods. This can not only improve the effectiveness of data, but also optimize data processing resources, and also provide a more reliable guarantee for computer program editing and network environment creation, and enhance the value of resource processing [5].

VI. Conclusion

All in all, in the current rapid development of computer technology, the popularity of computers is getting higher and higher. People's demand for computer engineering is also developing in the direction of diversification and enrichment. In order to promote the further development of computer engineering software construction, we need to pay attention to the effective application of database programming technology. Therefore, programmers need to master the problems that exist in the application of database programming technology. At the same time, they must also strengthen the analysis and research of database programming technology. Only if we fully master the database programming technology and improve the database programming level. During the application of database programming technology, programmers need to correctly understand the application advantages of the technology. Moreover, the application effect of database programming is also improved from the perspective of database storage mode selection, database performance, file encryption protection and programming development.

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